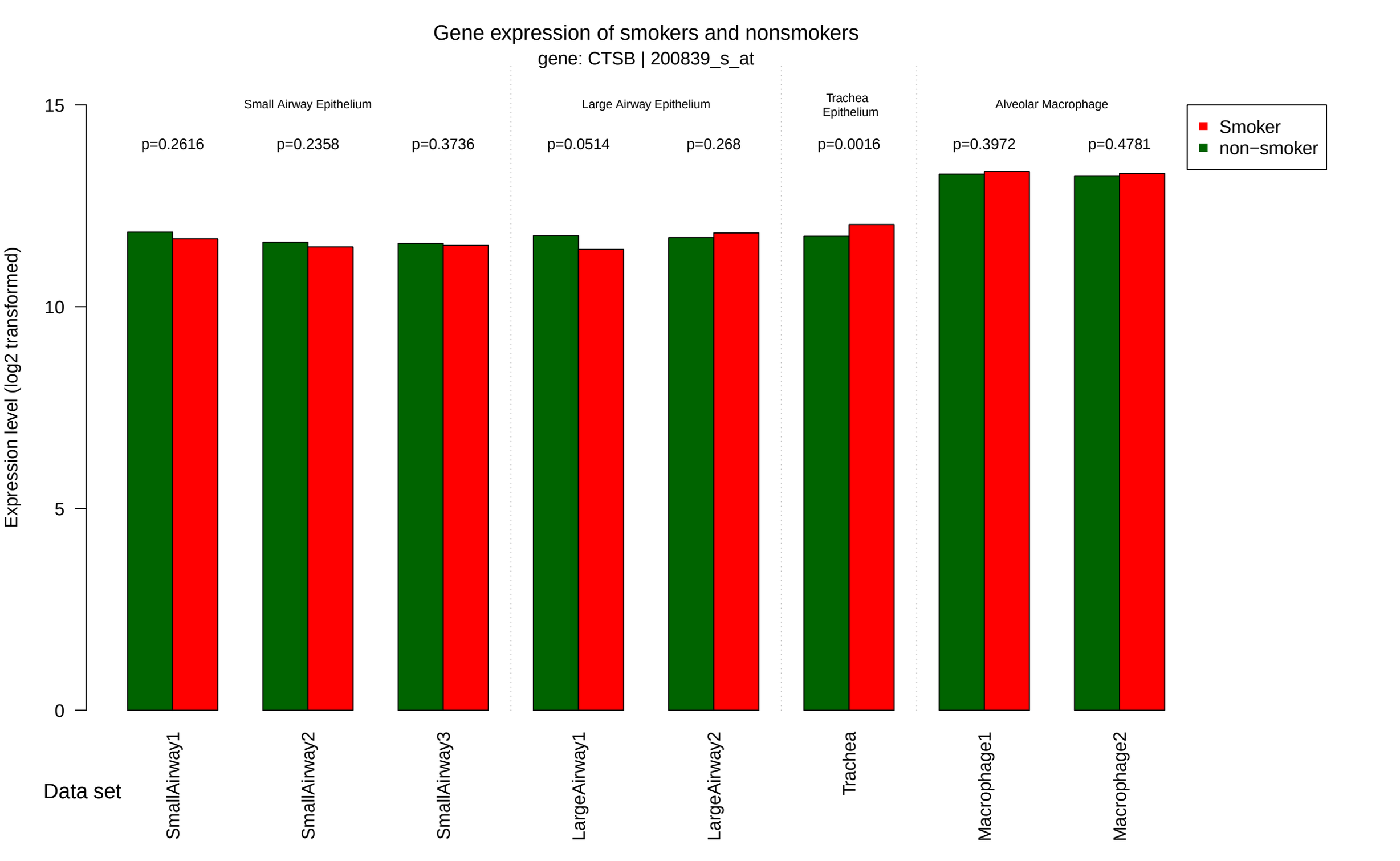
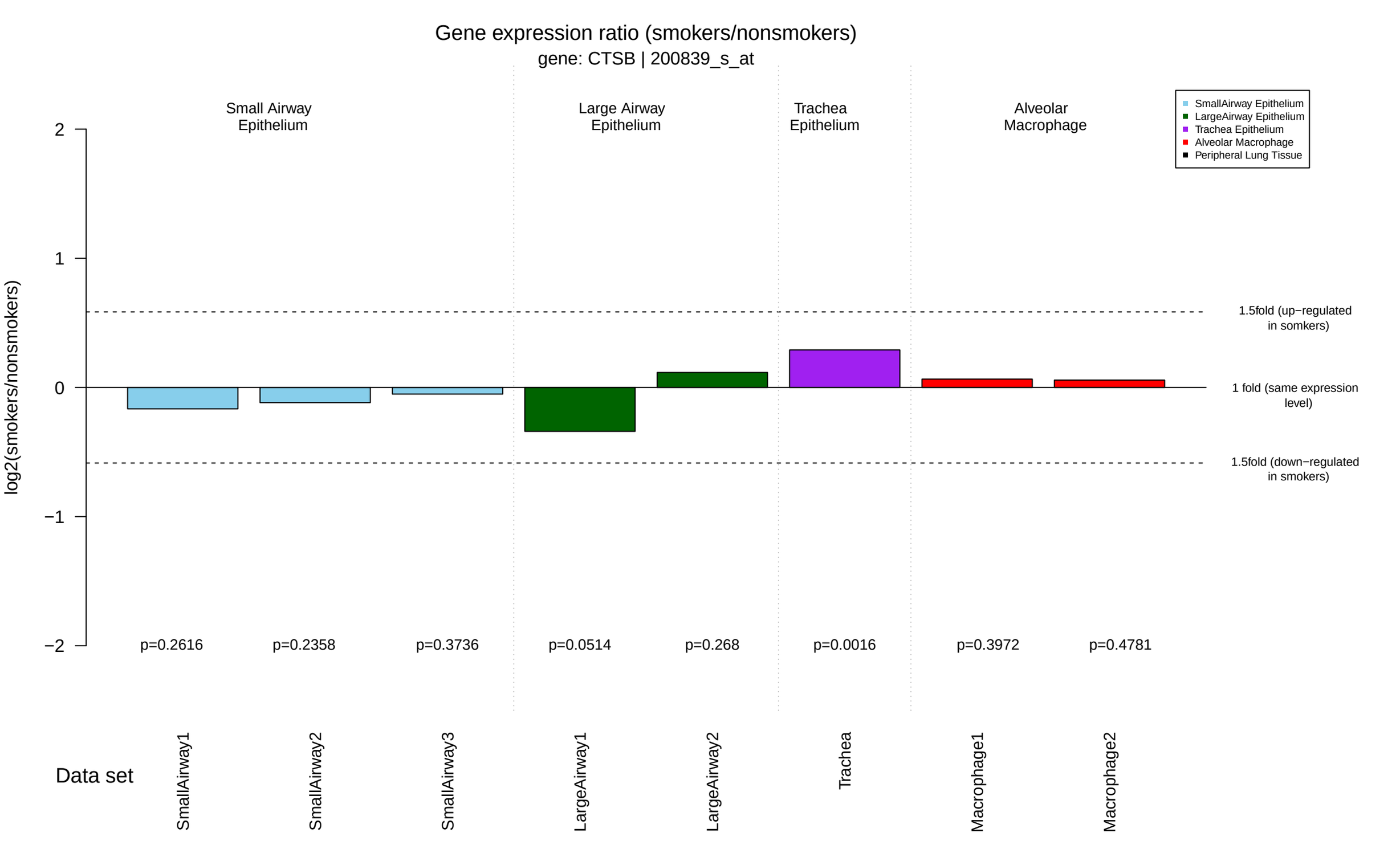
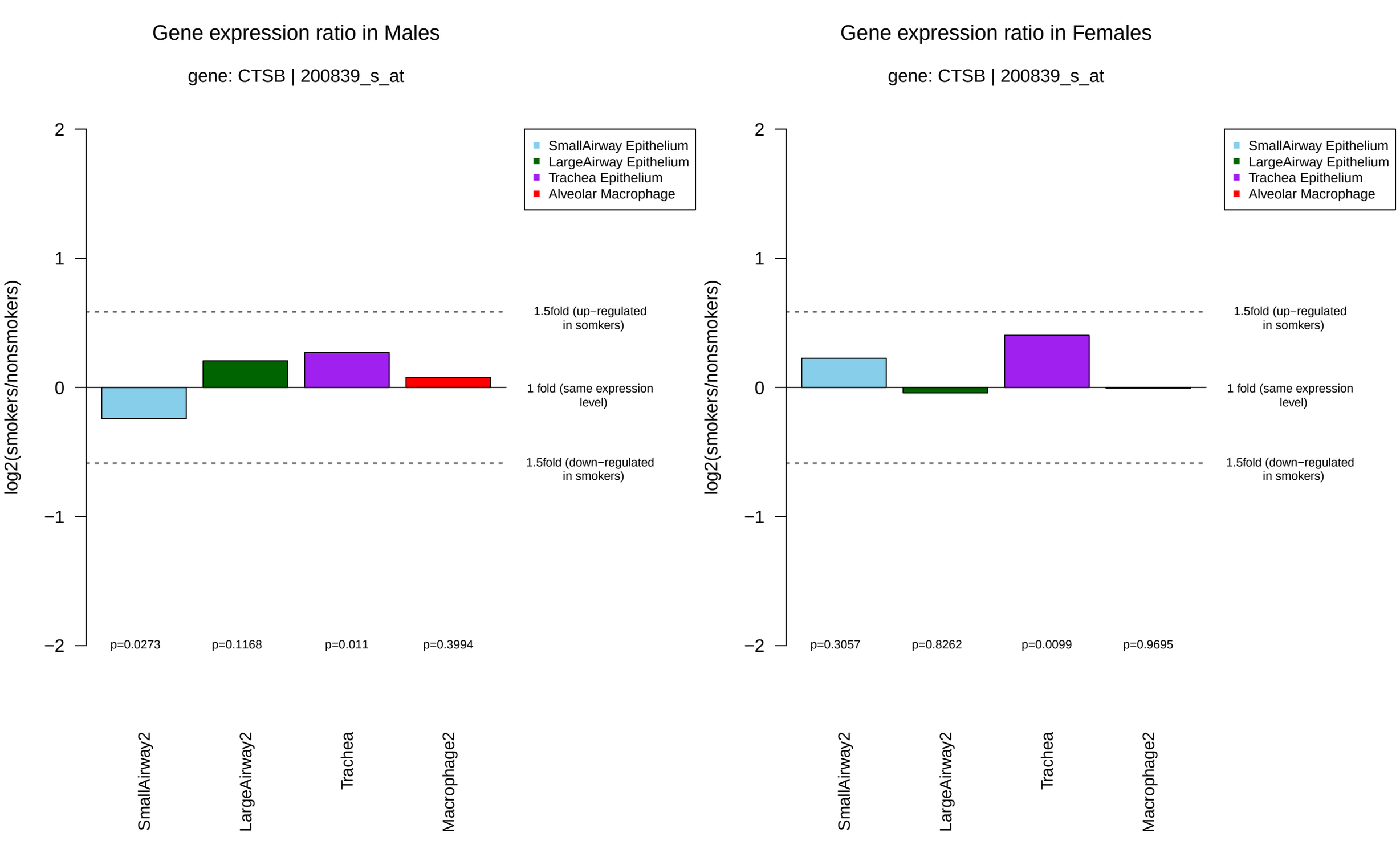
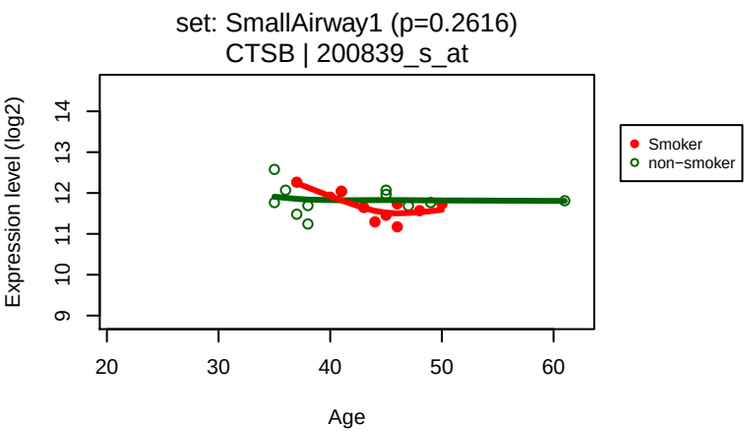
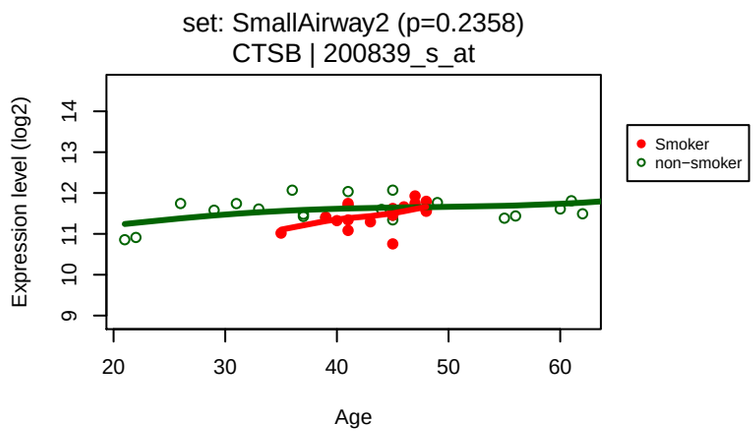
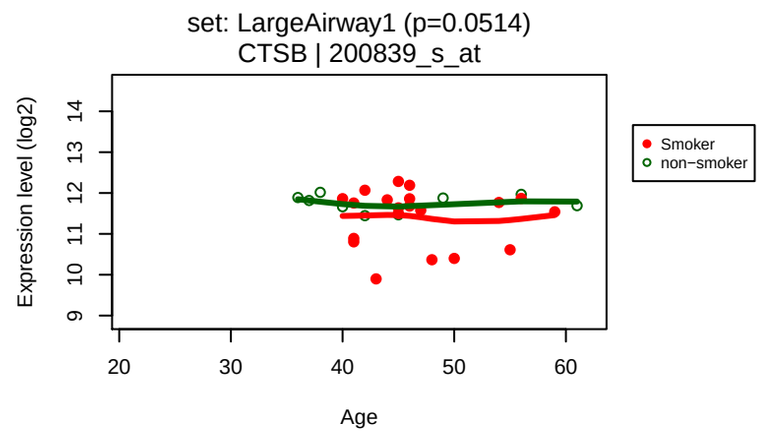

Gene expression of smokers and nonsmokers
gene: CTSB | 200839_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2616
p=0.2358
p=0.3736
p=0.0514
p=0.268
p=0.0016
p=0.3972
p=0.4781
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

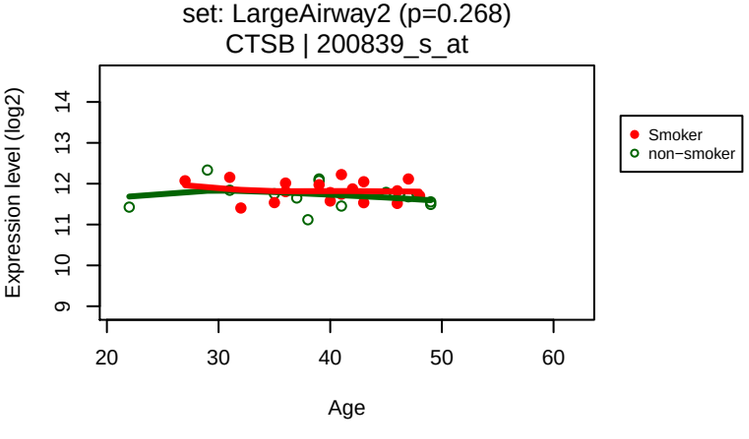

Gene expression ratio (smokers/nonsmokers)
gene: CTSB | 200839_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2616
p=0.2358
p=0.3736
p=0.0514
p=0.268
p=0.0016
p=0.3972
p=0.4781
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

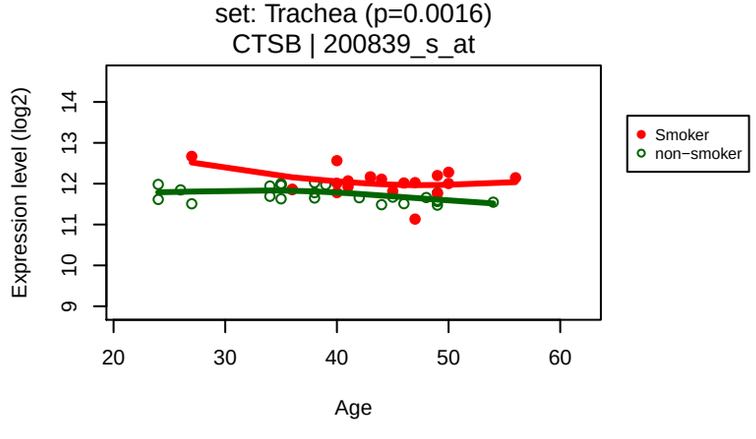

Gene expression ratio in Males
gene: CTSB | 200839_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0273
p=0.1168
p=0.011
p=0.3994
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CTSB | 200839_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3057
p=0.8262
p=0.0099
p=0.9695
SmallAirway2
LargeAirway2
Trachea
Macrophage2

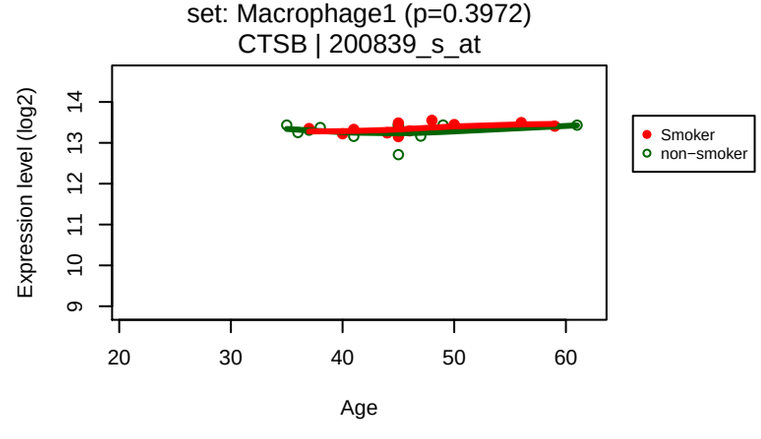

set: SmallAirway1 (p=0.2616)
CTSB | 200839_s_at
Expression level (log2)
Age
Smoker
non−smoker

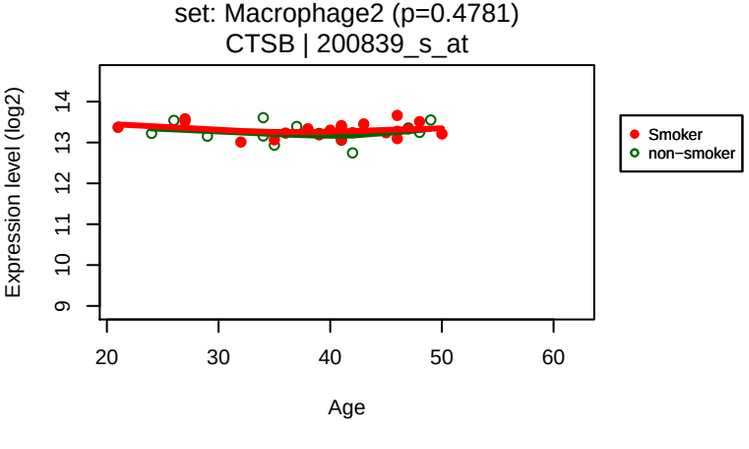

set: SmallAirway2 (p=0.2358)
CTSB | 200839_s_at
Expression level (log2)
Age
Smoker
non−smoker

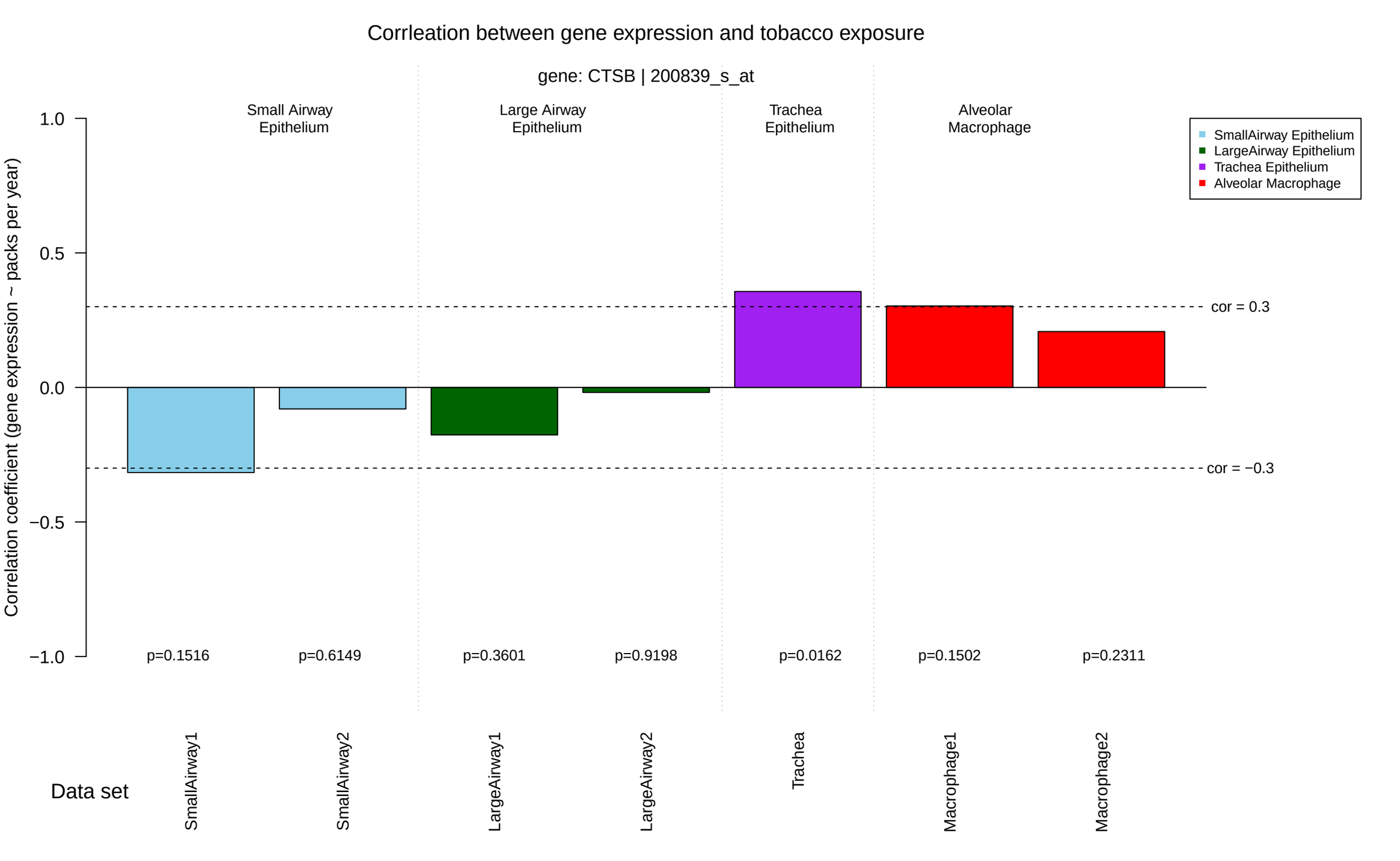

set: LargeAirway1 (p=0.0514)
CTSB | 200839_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.268)
CTSB | 200839_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0016)
CTSB | 200839_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3972)
CTSB | 200839_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4781)
CTSB | 200839_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CTSB | 200839_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1516
p=0.6149
p=0.3601
p=0.9198
p=0.0162
p=0.1502
p=0.2311
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2